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Q. Explain Software Processing Models in Detail.

Software Modelling :-

Defn: Software modelling is the process of creating abstract representations of a software system.
These models serve as blue print that guide developers, designers and stakeholders through system's structure, behaviour and functionality

(A) Waterfall Model :-

Requirements
Design
Implementation
(Testing) Verification
Maintanance

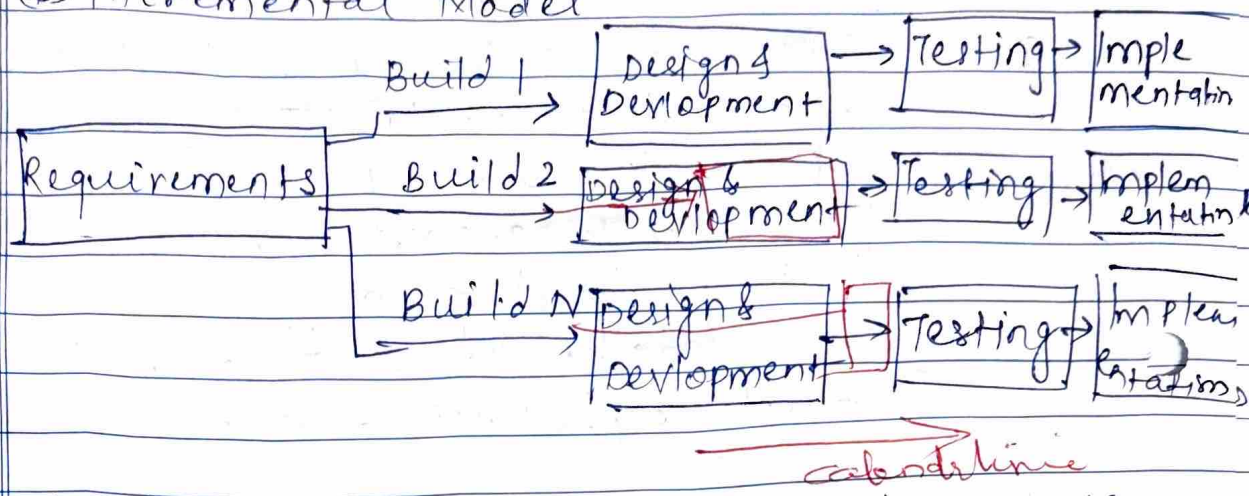
Waterfall model is famous and good version of SDLC for SE. The waterfall model is linear and sequential model, which means that development phase cannot begin until previous phase is completed.

Advantages: simple and easy to understand
well structured

Disadvantages: Not flexible to change
Late Testing phase increases risk

Best used when: Requirements are clear & stable
small and low risk project

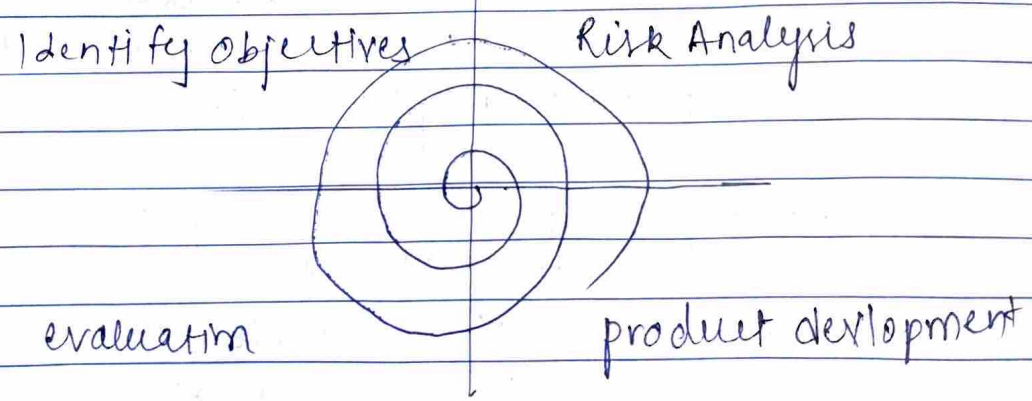
(B) Incremental Model




The incremental model develops the system in small parts (increments). ~~for~~ Each increment adds new functionality until the full system is completed

Adv:- Early delivery of partial system
Easier testing and debugging
Disadv: Requires good planning & design
Integration can be complex
Best used when: system can be divided into modules

(C) Spiral Model



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Spiral Model is a risk-driven software development model that combines itterative development with systematic risk management.

It has four phases:- Planning, Risk Analysis, Engineering and Evaluation

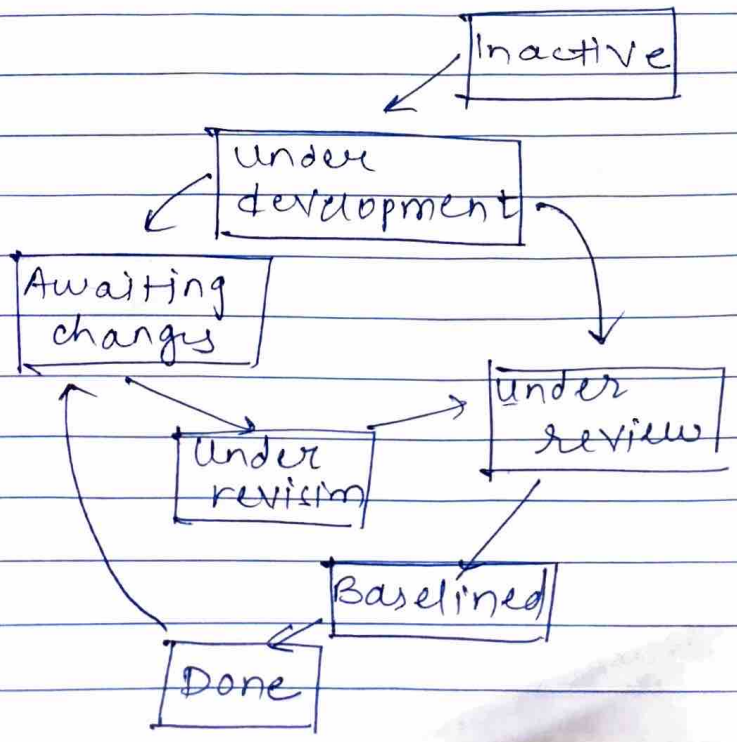
Planning → Define objects and alternatives
Risk Analysis → Identify and migitate risks
Engineering → Design, code and test
Evaluation → customer review and feedback

Adv:- early identification of risks
flexible and suitable for complex systems

DisAdv:- Expensive
Complex to Manage

Best used when: Large, high-risk projects
New technology is involved

(D) Concurrent Model:



Concurrent Devlopement model allows multiple phases of development to occur simultaneously

Working Principle:-
- Each activity exists in different states.
- Activities progress independently but are co-ordinated

Adv:- Faster development
Better Handling of changes
Suitable for parallel work

DisAdv: Complex to Manage
Requires effective communication

Best Used When:
Large project with multiple teams
Time to market is critical
System with frequent changes.